

# Recurrence does not control $\partial^2$ on $\ell^p$ when $p > 2$

Autonomous proof-discovery packet for arXiv:2605.21069

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**Status.** Candidate full counterexample and sharpened positive replacement; likely valid, human review requested. Model: GPT5.6.

## 1 Source question

Bartmann and Keller study the boundary operator on weighted, possibly infinite simplicial complexes and characterize  $\partial\partial = 0$  on  $\ell^2$  by recurrence of link graphs [1]. In the remark after their Theorem 4.1 (source PDF page 9), they prove the same implication on  $\ell^p$  for  $1 \leq p \leq 2$ , observe that their argument fails for  $p > 2$ , and ask for conditions governing the larger classes. The decisive excerpt is reproduced below.

which finishes the proof since  $\delta\pi^\rho u \in \ell^2$  as  $u \in \mathcal{D}_\rho$  by Lemma 3.1 (c).  $\square$

**Remark.** It is natural to ask for conditions when  $\partial\partial\omega(\rho) = 0$  for a larger class of functions  $\omega$ . In fact, it is not difficult to adapt the argument in Theorem 4.1 to show that the recurrence of all connected components of  $b_\rho$  implies that  $\partial\partial\omega(\rho) = 0$  for all  $\omega \in \ell^p$ ,  $p \in [1, 2]$ . The reason for that is the estimate

$$\|\delta\pi^\rho \varphi_n\|_q^q \leq 2^{q-2} Q^+(\pi^\rho \varphi_n)$$

for all  $2 \leq q < \infty$ . If  $1 < p \leq 2$ , then we apply this to the Hölder dual of  $p$ . If  $p = 1$ , then dominated convergence does the trick. For  $p > 2$  this argument does not work. It would be interesting to know if this is related to the existence of an  $\ell^p$ -Hodge decomposition or boundedness of Riesz-Transforms in  $\ell^p$ .

**Remark.** There are various criteria for recurrence of graphs. For an exhaustive treatment of the probabilistic aspects, one of the standard text books is [18]. For a more analytic approach, we refer to [19, Chapter 6] as [6] and the references therein.

The result below gives a negative answer to recurrence sufficiency for every  $p > 2$ . It also identifies the direct potential-theoretic replacement suggested by Hölder duality.

## 2 Definitions and proof intuition

For a fixed simplex  $\rho$ , write  $b_\rho$  for its weighted link graph. If  $1 < q < \infty$ , call a connected component  $X$  of this link  $q$ -parabolic when there are  $0 \leq \varphi_n \leq 1$  in  $C_c(X)$  such that  $\varphi_n \rightarrow 1$  pointwise and

$$\mathcal{E}_{q,\rho}(\varphi_n) := \frac{1}{2} \sum_{v,w \in X} b_\rho(v,w) |\varphi_n(v) - \varphi_n(w)|^q \longrightarrow 0.$$

This is the standard null-sequence formulation of graph  $q$ -parabolicity; see [2] for the general locally summable setting.

### Proof intuition

The source's recurrence proof tests the total boundary against cutoffs tending to one. For an  $\ell^p$  cochain, Hölder's inequality pairs it with the  $\ell^q$  norm of the cutoff gradient, where

$q = p/(p-1)$ . Thus the exact hypothesis is not small quadratic energy unless  $p = 2$ , but small  $q$ -energy. When  $p > 2$ , one has  $q < 2$ , and recurrence need not imply  $q$ -parabolicity.

The counterexample makes this mismatch visible as a conserved flux. Put conductance  $n$  on the  $n$ th edge of a half-line and value  $1/n$  on that edge. The weighted flux is identically one. Its divergence vanishes at every interior vertex but leaves one unit of charge at the endpoint. Quadratic resistance is  $\sum 1/n = \infty$ , so the link is recurrent; meanwhile the cochain's  $\ell^p$  cost is  $\sum n^{1-p} < \infty$  exactly when  $p > 2$ .

### 3 The positive replacement

**Proposition 1** (Dual parabolicity suffices). *Let  $1 < p < \infty$  and  $q = p/(p-1)$ . Fix  $\rho$  in a weighted simplicial complex in the sense of [1]. If every connected component of  $b_\rho$  is  $q$ -parabolic, then*

$$\partial\partial\omega(\rho) = 0 \quad (\omega \in D(\partial\partial) \cap \ell^p(\Sigma, m)).$$

*Proof.* Enumerate the components of  $b_\rho$ . By their  $q$ -parabolic null sequences and a diagonal choice, there are  $0 \leq \varphi_n \leq 1$  in  $C_c(\text{lk}(\rho))$  such that  $\varphi_n \rightarrow 1$  pointwise and  $\mathcal{E}_{q,\rho}(\varphi_n) \rightarrow 0$ . Indeed, on the first  $n$  components choose cutoffs with total  $q$ -energy at most  $1/n$ , and take their sum; monotone componentwise null sequences give pointwise convergence.

Use the localization maps  $\pi_\rho, \pi^\rho$  of [1]. Since  $\partial\omega \in D(\partial)$ , the function  $\pi_\rho\partial\omega$  is in  $\ell^1(\text{lk}(\rho), m_\rho)$ . Dominated convergence and the finite-support Stokes identity therefore give

$$\begin{aligned} m(\rho) |\partial\partial\omega(\rho)| &= \lim_{n \rightarrow \infty} \left| \sum_{\text{lk}(\rho)} m_\rho(\pi_\rho\partial\omega) \varphi_n \right| \\ &= \lim_{n \rightarrow \infty} \left| \sum_{\Sigma} m\omega \delta\pi^\rho \varphi_n \right| \\ &\leq \|\omega\|_p \lim_{n \rightarrow \infty} \|\delta\pi^\rho \varphi_n\|_q. \end{aligned}$$

For every simplex  $vv'\rho$  containing two link vertices, the only two terms in the localized coboundary differ by orientation and give  $\varphi_n(v) - \varphi_n(v')$ . Hence, exactly

$$\|\delta\pi^\rho \varphi_n\|_q^q = \frac{1}{2} \sum_{v,v'} b_\rho(v, v') |\varphi_n(v) - \varphi_n(v')|^q = \mathcal{E}_{q,\rho}(\varphi_n) \rightarrow 0.$$

The displayed boundary of boundary is therefore zero.  $\square$

### 4 A recurrent counterexample for every $p > 2$

**Theorem 1.** *There is one weighted one-dimensional simplicial complex  $(\Sigma, m)$  such that every link component is recurrent and, simultaneously for every  $p > 2$ , there is a single cochain*

$$\omega \in D(\partial\partial) \cap \ell^p(\Sigma, m) \quad \text{with} \quad \partial\partial\omega(\emptyset) = -1.$$

*Consequently, recurrence of the links does not imply the complex property on  $\ell^p$  for any  $p > 2$ .*

*Proof.* Let the vertices be  $v_0, v_1, \dots$  and let  $e_n = \{v_{n-1}, v_n\}$  for  $n \geq 1$ . Take the augmented one-dimensional complex

$$\Sigma = \{\emptyset\} \cup \{v_n : n \geq 0\} \cup \{e_n : n \geq 1\}$$

with weights

$$m(\emptyset) = 1, \quad m(v_n) = 2^{-n-1}, \quad m(e_n) = n.$$

This is locally summable in the source's sense:  $\sum_n m(v_n) = 1$  at the empty simplex, and each vertex has only one or two edge cofaces of finite weight.

Orient the augmented coboundary by

$$\theta(\emptyset, v_n) = 1, \quad \theta(v_{n-1}, e_n) = -1, \quad \theta(v_n, e_n) = 1.$$

Then  $\delta^2 = 0$ , since the two vertex contributions on each edge cancel. Define  $\omega$  to vanish on the empty simplex and all vertices and set

$$\omega(e_n) = \frac{1}{n} \quad (n \geq 1).$$

For every  $p > 2$ ,

$$\|\omega\|_p^p = \sum_{n=1}^{\infty} m(e_n) |\omega(e_n)|^p = \sum_{n=1}^{\infty} n^{1-p} < \infty.$$

All sums defining  $\partial\omega$  at vertices are finite. At the endpoint and the interior vertices, respectively,

$$\partial\omega(v_0) = \frac{-m(e_1)\omega(e_1)}{m(v_0)} = -2, \quad \partial\omega(v_n) = \frac{m(e_n)\omega(e_n) - m(e_{n+1})\omega(e_{n+1})}{m(v_n)} = 0 \quad (n \geq 1).$$

It vanishes on edges. Thus  $\partial\omega \in D(\partial)$ : the only nontrivial formal-domain sum is

$$\sum_{n \geq 0} m(v_n) |\partial\omega(v_n)| = 1.$$

Therefore  $\omega \in D(\partial\partial)$ , and

$$\partial\partial\omega(\emptyset) = \frac{1}{m(\emptyset)} \sum_{n \geq 0} m(v_n) \theta(\emptyset, v_n) \partial\omega(v_n) = -1.$$

It remains to check the link hypothesis. The link of  $\emptyset$  is the half-line with conductance  $b_{\emptyset}(v_{n-1}, v_n) = m(e_n) = n$ . Let  $H_N = \sum_{j=1}^N 1/j$  and define a finitely supported cutoff by

$$\eta_N(v_k) = \begin{cases} 1 - H_k/H_N, & 0 \leq k < N, \\ 0, & k \geq N, \end{cases} \quad H_0 = 0.$$

Then  $\eta_N \rightarrow 1$  pointwise and

$$\mathcal{E}_{2,\emptyset}(\eta_N) = \sum_{n=1}^N n \left( \frac{1}{nH_N} \right)^2 = \frac{1}{H_N} \rightarrow 0.$$

Thus this link is 2-parabolic, equivalently recurrent. Links of vertices have no link edges, so all their components are trivially recurrent; higher links are empty.

For completeness, this example also pinpoints Proposition 1. If  $q = p/(p-1) < 2$  and  $\phi \in C_c$  satisfies  $\phi(v_0) = 1$ , telescoping and weighted Hölder give

$$1 \leq \left( \sum_{n \geq 1} n |\phi(v_{n-1}) - \phi(v_n)|^q \right)^{1/q} \left( \sum_{n \geq 1} n^{-(p-1)} \right)^{1/p}.$$

The  $q$ -energy of such cutoffs is bounded away from zero. Hence the recurrent half-line is  $q$ -hyperbolic for precisely the dual exponents relevant to every  $p > 2$ .  $\square$

## 5 Scope, novelty, and review recommendation

Theorem 1 answers the source’s  $p > 2$  direction negatively if recurrence is retained as the proposed condition. Proposition 1 supplies a general sufficient condition but does not claim an unrestricted converse for arbitrary higher-dimensional links; extra faces can affect the global domain of a cochain built from a link flow. No  $\ell^p$ -Hodge decomposition or Riesz-transform equivalence is asserted.

The bounded novelty check (through 11 August 2026) searched all four local run indexes, the exact arXiv id and title, and combinations of *boundary of boundary*, *complex property*,  $\ell^p$ ,  $p > 2$ , *weighted half-line*, *unit flux*, and  $q$ -parabolicity. It found the source and the graph-parabolicity background [2], but no paper stating this counterexample or applying dual parabolicity to the source question. Because arXiv:2605.21069 is very recent, novelty is provisional.

Human review should focus on the iterated formal-domain check, the augmented orientation at  $\emptyset$ , the componentwise diagonal cutoff in Proposition 1, and a refreshed literature search.

## References

- [1] P. Bartmann and M. Keller, *The complex property of the boundary operator on simplicial complexes*, arXiv:2605.21069v1, 2026.
- [2] A. Adriani, F. Fischer, and A. G. Setti, *Characterizations of  $p$ -Parabolicity on Graphs*, arXiv:2507.13696v2, 2026.